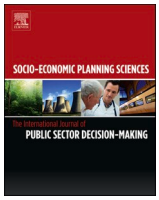
Socio-Economic Planning Sciences 98 (2025) 102146



Contents lists available at ScienceDirect

Socio-Economic Planning Sciences

journal homepage: www.elsevier.com/locate/seps





A hybrid optimization approach for designing sustainable intermodal freight transport under mixed uncertainty

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ARTICLE INFO

Keywords:
Intermodal freight transport
Sustainability
Resilience
Disruption
Uncertainty
Network data envelopment analysis

ABSTRACT

Intermodal freight transport stands out as an innovative logistics solution for combating climate change. However, the intricacies of planning resilient and sustainable intermodal freight transport systems necessitate further investigation. Addressing this critical research gap, this paper introduces a pioneering hybrid optimization framework tailored for designing a robust rail-road intermodal network that prioritizes sustainability while navigating disruptions and operational uncertainties. This framework adopts a multi-period approach to model intricate intermodal operations, including mode-switching activities. It strategically determines optimal locations for intermodal terminals and makes tactical decisions regarding inventory management and transport planning. The goal is to enhance the sustainability performance of intermodal terminals while simultaneously minimizing the overall total cost and carbon emissions of the network. The sustainability assessment of intermodal terminals employs a variant of network data envelopment analysis, considering a range of conflicting measures. Furthermore, the proposed optimization approach integrates P-robust and possibilistic chance-constraint programming to effectively mitigate disruption risks and operational uncertainties inherent in intermodal networks. To validate the efficacy of the proposed approach, two datasets are utilized, including a real-world case study of rail-road intermodal freight transport. The numerical findings showcase the significant benefits of this approach, demonstrating a remarkable 24 % reduction in maximum regret with a negligible 1.4 % increase in expected cost, highlighting its potential for sustainable and resilient intermodal freight transport systems. This research provides valuable insights for freight transport companies to enhance resilience against disruptions and manage operational uncertainties, while also contributing to societal goals of sustainable development through reduced environmental impacts.

1. Introduction

Our modern society heavily relies on freight transportation [1]. However, the rapid expansion of freight transportation has raised significant environmental concerns [2]. The transportation sector now stands as the second-largest contributor to CO2 emissions globally [3], accounting for over 20 % of energy-related emissions in 2023 [4]. In the European Union, inefficiencies in freight transport—such as 40–60 % empty haulage rates—along with insufficient risk and disruption management, have amplified ecological and social challenges. These issues highlight the urgent need for sustainable practices in freight operations [5].

One promising sustainable logistics solution that can help reduce environmental carbon footprint while meeting growing demands is multi-modal freight transport [6]. Multi-modal transportation integrates various modes of transport, including road, rail, and maritime, into a cohesive network that seamlessly coordinates the movement of goods from their origin to their destination [7,8]. This approach leverages greener modes of transport, specifically rail and maritime, contributing to environmental sustainability [9]. Intermodal transport is a specific type of multi-modal transport [10] where operations at freight transfer points, called intermodal terminals, keep freight in the same loading unit throughout the journey [11]. Factors like environmental considerations, streamlined tracking capabilities, and reduced transportation

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https://doi.org/10.1016/j.seps.2024.102146
Received 26 May 2024; Received in revised form 7 December 2024; Accepted 20 December 2024
Available online 7 January 2025
0038-0121/© 2024 Published by Elsevier Ltd.

costs have all contributed to the growing significance of intermodal transportation, establishing it as a vital component of future transportation systems [12].

However, designing an intermodal transport network is a complex endeavor that involves navigating various challenges. Firstly, selecting suitable locations for terminal establishment is a multifaceted and multi-criteria decision-making process due to conflicting indicators that influence this decision [13,14]. The strategic positioning of these intermodal terminals is vital for facilitating efficient transportation, handling, storage, and transshipment of goods [15]. Consequently, the choice of terminal locations can have significant economic, environmental, and social impacts, thereby playing a crucial role in determining the overall sustainability performance of supply chains [16]. From a technical standpoint, designing intermodal transport systems presents greater complexities compared to unimodal transport systems. This complexity arises from the need for extensive coordination and planning among different modes of transport [17]. For instance, it requires detailed modeling of mode-switching operations at terminals to ensure smooth transitions between different modes of transport. Additionally, addressing technical intricacies involves determining optimal inventory levels at terminals and managing the quantity of transport from each mode to avoid any logistical imbalances at the destination [18].

Furthermore, intermodal transport networks face a higher susceptibility to disruptions [19,20], including extreme weather events [21], unplanned upstream network delays, equipment failures, labor actions [22], and inefficiencies within rail yards [23]. These disruptions can lead to a significant and prolonged economic downturn and a drastic decrease in efficiency within freight transportation networks [24]. For instance, the devastating earthquake in California in 1994 resulted in a loss of nearly $1.5 billion in the transportation sector due to network destruction [25]. Similarly, natural disasters like the Mississippi River flood and storms in New York and New Jersey during the 2010s led to billions of dollars in financial losses and disruptions in major transportation networks. Human-related factors also contribute to disruptions, as evidenced by the $650 billion financial loss incurred during a workers' strike in Port Beach, California, in 2012 [26]. In addition to these disruptive events, operational uncertainties such as supply and demand fluctuations pose significant challenges that must be considered in decision-making processes [27]. These uncertainties add another layer of complexity to managing intermodal transport networks effectively.

This research addresses these critical challenges by posing three key questions.

- How can an efficient and environmentally sustainable rail-road intermodal transport system be designed?
- How can the resilience of intermodal network be improved to withstand random disruptions?
- How can the operational uncertainties inherent in the system be effectively managed?

To answer these questions, an innovative hybrid optimization framework is proposed for a sustainable rail-road intermodal freight system that is resilient to disruptions and operational uncertainties. This framework leverages a modified version of network data envelopment analysis (NDEA) to initially assess the sustainability of intermodal transport, considering a range of conflicting criteria. Subsequently, we employ a multi-period bi-objective mixed-integer programming (MIP) model to capture the intricacies of intermodal operations, including mode-switching and inventory planning. By integrating P-robust and possibilistic chance-constraint programming (PCCP), our approach effectively addresses disruption risks and inherent operational uncertainties within the network. This comprehensive methodology enables strategic decision-making regarding the optimal locations of intermodal terminals, as well as tactical planning related to inventory management and transport logistics. Our primary goal is to enhance sustainability metrics at intermodal terminals while simultaneously reducing the overall total cost and carbon emissions of the network. To validate the efficacy of our proposed approach, we conduct a detailed analysis using a real-world case study of rail-road intermodal freight transport as well as a data set from the literature. Through this analysis, we derive valuable managerial insights that contribute significantly to bridging existing research gaps in this domain. As a result, this study makes significant contributions by.

- Developing a multi-period, multi-objective planning approach for the sustainable redesign of rail-road intermodal systems, encompassing mode-switching operations and inventory management.
- Evaluating the sustainability performance of intermodal terminals based on economic, environmental, and social indicators.
- Addressing random disruption risks and operational uncertainties within the intermodal network, including demand and supply considerations.
- Introducing a novel optimization framework that integrates cutting-edge decision-making approaches such as NDEA, bi-objective MIP, P-robust, and PCCP, offering a timely and effective solution to complex challenges.

The remainder of this paper is organized as follows: Section 2 examines the relevant literature. Section 3 lays out the problem statement and assumptions. Our proposed methodology is detailed in Section 4. Section 5 encompasses the case study, computational results, and managerial insights. Lastly, Section 6 concludes the paper and offers directions for future research.

2. Literature Review

Given the critical role of facility location decisions in intermodal freight transport planning, numerous studies have developed optimization models for determining the optimal locations of intermodal networks. For instance, Fotuhi and Huynh [28] formulated a MIP approach to address the rail-road intermodal expansion problem amidst uncertainties in supply and demand, alongside disruptions in terminals and links. Their strategies encompassed retrofitting critical rail links, establishing new terminals, and expanding existing ones. Likewise, Fotuhi and Huynh [29] devised a mathematical model to tackle the reliable multi-period intermodal freight network expansion problem, employing a Simulated Annealing algorithm for resolution. W. Zhang et al. [30] developed a tri-objective model targeting cost, time, and reliability optimization for hub locations within an intermodal freight network. K. Yang et al. [31] also contributed by developing a fuzzy possibility model to enhance reliability through the establishment of intermodal terminals, with the model solved using Tabu Search. However, sustainability criteria were not integrated into these optimization models.

Research efforts have also been directed towards exploring the environmental dimensions of inter/multimodal freight transportation networks. For instance, Agbo and Zhang [32] developed mathematical models to address sustainability challenges such as environmental pollution and road congestion in synchromodal freight transport systems. Their numerical results demonstrated that implementing synchromodality concepts has significant potential to reduce total transportation and service costs while achieving substantial environmental benefits, including lower road transport emissions and reduced congestion. Torres et al. [33] evaluated the potential of rail-road intermodal operations in mitigating climate change impacts. Kelle et al. [34] suggested a simulation model for an intermodal freight transport system, evaluating the trade-off between environmental and other objectives. They conducted a what-if analysis to assess the advantages of mode-changing initiatives in the Louisiana freight network. Given the conflicting interests of different stakeholders and ensuring sustainability development indexes, Tadić et al. [35] devised a mechanism to prioritize

Table 1
Overview of relevant literature.

References	Strategic Decisions for Terminal Location	Model Features			Uncertainty Approach				Multi-Objective	Multi-Period	Disruption Modeling	Sustainability Evaluation of terminals	Research Methodology		
		Inventory Planning	Mode-Switching Planning	Epistemic Uncertainty Modeling	Deterministic	Fuzzy/ Possibilistic	Stochastic	Robust					NDEA	P-robust	PCCP
Uddin and Huynh [59]							×				×				
Agbo and Zhang [32]					×							×			
Mohammadi et al. [60]	×			×		×	×		×		×				×
Marufuzzaman and Ekşioğ; lu [61]		×					×			×	×				
Fazayeli et al. [62]	×		×	×		×									
Maiyar and Thakkar [50]	×					×				×	×	×			
Wolfinger et al. [63]					×										
Uddin and Huynh [51]				×		×					×				
Demir et al. [36]			×		×				×						
D. Zhang et al. [64]							×								
Guo et al. [65]		×					×			×					
Z. Yang et al. [66]					×										
Ziaei and Jabbarzadeh [27]				×				×	×						
Gao et al. [67]							×		×		×				
Ke and Verma [55]							×				×				
Hrušovský et al. [54]							×		×		×				
Okyere et al. [68]			×		×										
Badyala et al. [56]	×	×					×			×	×				
Wang and Li [69]	×				×							×			
Song et al. [70]			×		×				×						
Rodríguez-Espíndola et al. [44]	×	×	×				×		×						
Badyala et al. [56]	×						×			×	×				
Akyüz et al. [57]					×					×	×				
Gandhi et al. [45]						×						×			
Gandhi et al. [46]						×						×			
Somsai et al. [71]					×				×						
Shoukat [72]		×			×				×	×					
Temizceri and Kara [47]					×				×			×			
Ferjani et al. [73]			×		×				×						
Goodarzi et al. [10]							×				×				
Nayak et al. [74]		×			×					×					

(*continued on next page*)

Table 1 (*continued*)

References	Strategic Decisions for Terminal Location	Model Features			Uncertainty Approach				Multi-Objective	Multi-Period	Disruption Modeling	Sustainability Evaluation of terminals	Research Methodology		
		Inventory Planning	Mode-Switching Planning	Epistemic Uncertainty Modeling	Deterministic	Fuzzy/ Possibilistic	Stochastic	Robust					NDEA	P-robust	PCCP
Tafur and Padgett [58]							×				×				
Goodarzi et al. [75]							×				×				
This work	×	×	×	×		×	×	×	×	×	×	×	×	×	×

the features concerned with the intermodal terminal's development. To resolve the problem, they developed a hybrid MCDM model in the fuzzy environment. Demir et al. [36] crafted a bi-objective model for a green intermodal network, focusing on minimizing the overall transportation and greenhouse gas (GHG) emissions-related costs. Kumar and Anbanandam [37] utilized a Grey-DEMATEL technique to identify the causal relationships between 22 inhibitors and the strength of their interrelationships in the Indian railroad intermodal context. Heinold and Meisel [38] crafted a multi-objective formulation for optimal freight routing in intermodal rail-road networks, establishing emission limits to control greenhouse gases. Kumar and Anbanandam [39] identified twenty-five sub-criteria including social, economic, environmental, political, and technical dimensions for the sustainability assessment of the multimodal freight terminal's location. The significance of these indexes is calculated using the Intuitionistic Fuzzy-Analytic Hierarchy Process technique.

Tamannaei et al. [40] developed a game theoretic framework where the government as the leader imposes a tax on fuel usage to facilitate the modal shift of shippers from the sole road to road-rail intermodal systems. Similarly, Taheri and Tamannaei [41] presented a mathematical model for a multi-level intermodal network design problem, where governments on the upper level act as the leader and freight shippers on the lower level, as the follower. Mohri and Thompson [42] evaluated the viability of a short-term policy, named controlled tariff discounting policy in a sustainable intermodal freight transport network design problem, where the government positions the intermodal hubs. In rail-road freight transport systems, Fulzele and Shankar [43] identified 24 sustainability indicators and classified them into four categories. They benefited from a hybrid technique, intuitionistic fuzzy numbers with the graph theoretic and matrix approach, to quantify and prioritize the determined sustainability practices from a panel of freight transport experts in India. For humanitarian logistics, Rodríguez-Espíndola et al. [44] proposed a two-stage stochastic model for a sustainable intermodal network. Their model optimized facility location, resource allocation, intermodal operations, and distribution activities while minimizing costs, relief shortages, and GHG emissions. Gandhi et al. [45] introduced a multi-step mechanism to prioritize modal shift solutions to rising risks from leveraging the rail-road intermodal transport system. Likewise, applying an MCDM framework, Gandhi et al. [46] ranked the benefits derived from employing various modal shift enablers facilitating rail-road intermodal freight transport adoption, promoting sustainability indicators. Developing a bi-objective model (profit maximization and GHG emissions minimization), Temizceri and Kara [47] analyzed a green rail-water-road intermodal freight system in Turkey. They used different machine learning algorithms to predict the volume of GHG emissions. Nonetheless, these studies did not factor in disruption risks.

Several studies have delved into disruptions within inter/multimodal transportation systems. Chen et al. [48] developed optimization models to manage operations in rail-road intermodal networks under disruptive scenarios. Sarhadi et al. [49] devised an analytical framework to safeguard rail-road intermodal freight transportation networks from disruptions, adopting fortification plans to mitigate worst-case attacks. Maiyar and Thakkar [50] presented a rail-road intermodal freight transportation model for food grain shipping in disrupted hubs, focusing on total cost optimization through multi-period mixed-integer nonlinear optimization. Uddin and Huynh [51] devised a mathematical model for a rail-road multi-modal freight transportation network facing capacity degradation in transportation modes. For hazardous materials shipping via rail-road intermodal transportation networks, Ke [52] introduced a scenario-based robust optimization model accounting for random disruptions in terminals. Kramarz et al. [53] analyzed the effect of disruption in freight flows in the rail transport system as the core of the intermodal freight transport network. They quantified the reliability by calculating the number of cancelled trains and the average delay time of trains. Given the disruption impact on intermodal networks, Hrušovský et al. [54] presented a hybrid optimization-simulation framework,

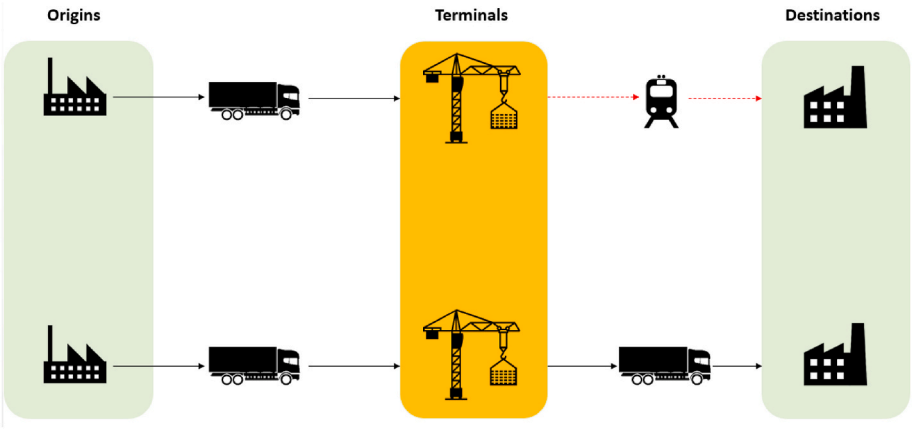


Fig. 1. Rail-road intermodal transport network under investigation.

combining offline and online re-planning strategies (e.g., alternative routes) once an expected event occurs.

Ke and Verma [55] proposed a framework combining optimization and regression analysis for rail-road intermodal transportation network recovery from random disruptions, developing mixed-integer formulations for normal and post-disruption operations. Badyala et al. [56] crafted a two-stage stochastic programming model for intermodal facility location and freight distribution, factoring in the likelihood of random disruptions at shipper facilities and/or intermodal terminals. However, these works did not address epistemic uncertainty and sustainability concerns. Akyüz et al. [57] analyzed the partial and complete replanning in an intermodal freight transport system exposed to various sources of disruption, including delays and order cancellations. They leveraged a column generation technique to navigate the two investigated network configurations. Nonetheless, their work lacks the terminal spots assessment and operations within, such as the mode-switching process and inventory management. Tafur and Padgett [58] examined a flow-based port capacity model for a maritime-rail-road intermodal freight network susceptible to storm disruption, targeting the resilience assessment of the investigated network. Goodarzi et al. [10] introduced a two-stage stochastic model for an intermodal freight transport system, emphasizing environmental and social sustainability metrics such as GHG emissions, traffic congestion, and noise pollution. Their framework evaluated the benefits of consolidation strategies and enhanced the network's resilience to unexpected disruptions by incorporating preparedness and recovery measures.

Table 1 provides a summary of relevant literature, focusing on mixed-integer optimization models and fuzzy programming approaches for designing and planning intermodal freight transport systems, particularly those addressing sustainability challenges and disruption risks.

An analysis of the literature underscores several research gaps that this study aims to address.

- Existing studies predominantly focus on strategic-level intermodal transport planning, neglecting crucial tactical and operational aspects such as mode-switching planning, inventory management, and multi-period transport planning.
- While some research has explored carbon emissions in transport modes, there is a dearth of studies evaluating broader sustainability measures, particularly those relevant to intermodal terminals.
- Disruption risks in intermodal transportation have received attention, but research addressing the epistemic nature of operational uncertainties like demand and supply variations is limited.
- Methodologically, the development of a hybrid optimization framework that harnesses the strengths of NDEA, P-robust, and PCCP, while remaining time-efficient for solving a multi-objective model in a real rail-road intermodal case study, is notably scarce in the current literature.

3. Problem statement and assumptions

The problem is to develop a sustainable intermodal transport system for transporting commodities from manufacturing sites (origins) to various demand points (destinations) across multiple time periods, utilizing a combination of rail and road transport modes. As illustrated in Fig. 1, the terminals within this intermodal network serve as intermediate points where mode-switching operations occur. This system is vulnerable to disruptions that can render terminals dysfunctional and limit available transport modes. Adding to this complexity is the epistemic uncertainty in both demand and supply.

To effectively address these disruptions and operational uncertainties, we implement two critical resilience strategies (for a comprehensive overview of resilience strategies, see Refs. [76–78]. The first strategy is a rerouting scheme that provides alternative routes when disruptions make intermodal terminals unusable, thereby preventing inefficient freight transport [48,54]. By selecting alternative routes, we can mitigate the impact of these disruptions on the overall supply chain. The second strategy involves maintaining buffer inventory to alleviate the negative consequences of random disruptions. While this approach incurs additional costs, it ultimately helps avoid higher expenses related to customer dissatisfaction and ensures a high level of service [76,79,80]. Furthermore, our optimization approach incorporates a resilience strategy during the terminal location stage, taking into account the risk of severe weather conditions. This proactive measure prevents the selection of locations prone to extreme weather events, such as heavy snowfall or prolonged storms [26].

The overarching objective is to optimize the sustainability performance of intermodal terminals while simultaneously minimizing the overall total cost and GHG emissions of the network. The sustainability performance of intermodal terminals is evaluated based on various measures such as their impact on the ecological landscape, proximity to residential districts, and vulnerability to adverse meteorological condition.

The initial assumptions for this study are as follows.

- Existing terminals are already operational, and there is flexibility to incorporate new terminals into the current network.
- The system allows for lost customer demand, which can occur due to capacity constraints at terminals or links during disruptions or normal operational phases.
- Mode switching is restricted exclusively to terminals within the network.

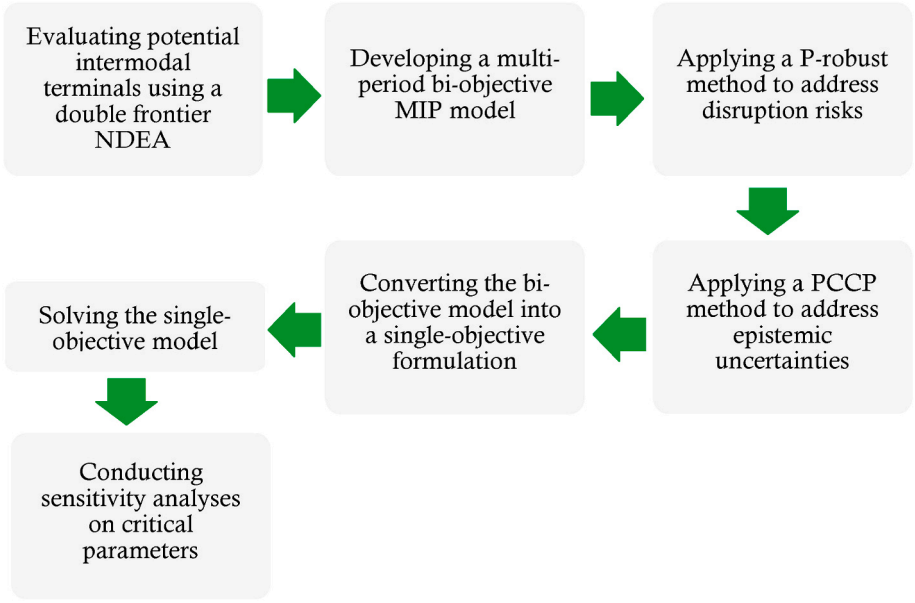


Fig. 2. Steps of the proposed methodology.

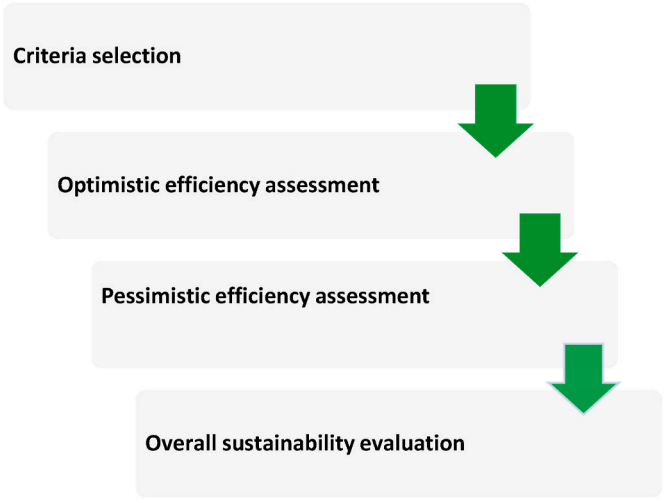


Fig. 3. Steps of double frontier NDEA method.

- The capacity of terminals and links between nodes is limited by design.
- Storage capacity at terminals is also limited.
- The problem focuses on a single-commodity transport system.

The primary decisions that need to be made include.

- Determining the optimal number and strategic locations for new intermodal terminals within the network.
- Establishing the quantity of freight to be transported via each transport mode under various disruption scenarios.
- Calculating the optimal inventory levels required at each terminal during different time periods to maintain operational efficiency.

4. Proposed methodology

To design a sustainable and resilient intermodal transport system as described in Section 3, we propose a novel hybrid optimization methodology. The first phase involves conducting a sustainability evaluation for candidate intermodal terminals based on economic, environmental, and social sustainability measures using a variant of NDEA. In the next phase, we develop a bi-objective MIP model to capture the intricacies of intermodal operations, such as mode-switching and inventory planning across different time periods. This multi-period model is then expanded using P-robust and PCCP approaches in subsequent steps to address disruption risks and epistemic uncertainties in supply and demand parameters. Using the TH method, introduced by Torabi and Hassini [81], we convert the resulting model into a single-objective formulation for solving. The final step includes conducting sensitivity analysis to gain managerial insights.

Fig. 2 outlines these steps. In the following sections, we provide detailed explanations and justifications for each step.

4.1. Sustainability evaluation of intermodal terminals using double frontier NDEA

The selection of optimal sites for new terminals significantly influences the sustainability performance of the intermodal freight transportation network. This selection process involves navigating numerous conflicting sustainability criteria. The literature extensively studies sustainability indicators for assessing the performance of freight transportation in both rail and road sectors (see Ref. [8] for a comprehensive review). To determine the relevant criteria, a committee of experts utilizes the triple bottom line approach, which emphasizes economic, environmental, and social dimensions to evaluate the performance of a sustainable intermodal network [82].

Once the criteria are established, a double frontier NDEA approach is employed to evaluate candidate locations based on various sustainability metrics for both rail and road modes. Fig. 3 illustrates the steps involved in this method.

The rationale behind using the proposed double frontier NDEA is that it overcomes the limitations of traditional DEA methods, which often overlook the internal connections between criteria and treat each Decision-making Unit (DMU) as a black box [83]. Unlike typical DEA methods that tend to evaluate candidates with an optimistic bias, double frontier NDEA provides a more precise evaluation by considering both optimistic and pessimistic viewpoints [84].

The following sections introduce the mathematical notation used in the optimistic and pessimistic models.

Sets.

J	Sets of DMUs (i.e., terminals) ($j, o \in J$)
A	Sets of input indicators ($a \in A$)
B	Sets of rail output indicators ($b \in B$)
C	Sets of road output indicators ($c \in C$)
D	Sets of joint output indicators ($d \in D$)

Parameters.

x_{aj}	The values of input indicator $a \in A$ for DMU $j \in J$
y_{bj}^{rail}	The values of rail output indicator $b \in B$ for DMU $j \in J$
y_{cj}^{road}	The values of road output indicator $c \in C$ for DMU $j \in J$
y_{dj}^{joint}	The values of joint output indicator $d \in D$ for DMU $j \in J$

Decision variables.

δ_a	The assigned weight to input indicator a
Ψ_b	The assigned weight to rail output indicator $b \in B$
χ_c	The assigned weight to road output indicator $c \in C$
λ_d	The assigned weight to joint output indicator $d \in D$
α_j	Optimistic value for each DMU $j \in J$ (See Equation (1))
β_j	Pessimistic value for each DMU $j \in J$ (See Equation (7))
σ_o	Overall score for each DMU $o \in J$ (See Equation (13))

It should be noted that the optimal values for α_j and β_j for each DMU $j \in J$ are denoted by α_j^* and β_j^*, respectively. Here, the indices $o \in J$ and $j \in J$ are used interchangeably to represent different DMUs within the equations. Therefore, α_o^* and β_o^* specifically denote the optimal values of α_o and β_o for each DMU $o \in J$. In this context, inputs and outputs are defined by sustainability criteria: a criterion is classified as an output when a higher value is more desirable and as an input when a higher value is less desirable.

The following subsections provide a concise overview of the optimistic and pessimistic models, along with the overall evaluation measure used in the double frontier NDEA approach. For a comprehensive explanation of the double frontier NDEA methodology, refer to Khanjarpanah et al. [85], and for applications of DEA in the transport industry, see Mahmoudi et al. [86].

4.1.1. Optimistic model

The optimistic model assesses the best-case sustainability performance of each terminal, estimating the highest efficiency achievable under ideal conditions. This approach sets an ideal benchmark, highlighting a terminal's maximum potential sustainability. The model thus aims to maximize the efficiency (i.e., $\frac{Output}{Input}$) of each DMU by assuming ideal conditions, effectively establishing an upper bound on sustainability performance.

Based on Khanjarpanah et al. [85], the optimistic model for each DMU $o \in J$ can be formulated as follows:

$$\alpha_o = Maximize\ \frac{1}{2}\left(\sum_{b\in B}\Psi_b y_{bo}^{rail} + 2\sum_{d\in D}\lambda_d y_{do}^{joint} + \sum_{c\in C}\chi_c y_{co}^{road}\right) \tag{1}$$

$$s.t.\ \sum_{a\in A}\delta_a x_{ao} = 1 \tag{2}$$

$$\sum_{b\in B}\Psi_b y_{bo}^{rail} + 2\sum_{d\in D}\lambda_d y_{do}^{joint} + \sum_{c\in C}\chi_c y_{co}^{road} - 2\sum_{a\in A}\delta_a x_{aj} \le 0, \forall j \in J \tag{3}$$

$$\sum_{b\in B}\Psi_b y_{bj}^{rail} + \sum_{d\in D}\lambda_d y_{do}^{joint} - \sum_{a\in A}\delta_a x_{aj} \le 0, \forall j \in J \tag{4}$$

$$\sum_{c\in C}\chi_c y_{co}^{road} + \sum_{d\in D}\lambda_d y_{do}^{joint} - \sum_{a\in A}\delta_a x_{aj} \le 0, \forall j \in J \tag{5}$$

$$\delta_a, \Psi_b, \chi_c, \lambda_d \ge 0, \forall\ a \in A, b \in B, c \in C, d \in D \tag{6}$$

Equation (1) calculates the model's optimal weighted output values, where the weighted input values are kept constant at 1, as specified in Equation (2). Equations (3)–(5) impose that, for each DMU, the efficiency (i.e., $\frac{output}{input}$) does not exceed 1. More specifically, Equation (3) applies to joint indicators (covering both rail and road), Equation (4) addresses only rail indicators, and Equation (5) focuses solely on road indicators. The constraints for the decision variables are controlled by equation (6), ensuring these variables are non-negative. To achieve meaningful results, the number of DMUs should exceed the combined total of inputs and outputs, following the guidelines of [87].

The optimistic model is solved individually for each DMU $o \in J$, and the optimal efficiency value for each DMU $o \in J$, denoted as α_o^*, is obtained.

4.1.2. Pessimistic model

The pessimistic model serves as the counterpoint to the optimistic model for DMUs. This approach calculates minimum efficiency, effectively capturing a worst-case scenario for the sustainability performance of terminals. It operates under the assumption that each DMU must produce more output than it consumes, even in less favorable conditions [83].

According to Khanjarpanah et al. [85], the pessimistic model for each DMU $o \in J$ can be formulated as follows:

$$\beta_o = minimize\ \frac{1}{2}\left(\sum_{b\in B}\Psi_b y_{bo}^{rail} + 2\sum_{d\in D}\lambda_d y_{do}^{joint} + \sum_{c\in C}\chi_c y_{co}^{road}\right) \tag{7}$$

$$s.t.\ \sum_{a\in A}\delta_a x_{ao} = 1 \tag{8}$$

$$\sum_{b\in B}\Psi_b y_{bo}^{rail} + 2\sum_{d\in D}\lambda_d y_{do}^{joint} + \sum_{c\in C}\chi_c y_{co}^{road} - 2\sum_{a\in A}\delta_a x_{aj} \ge 0, \forall j \in J \tag{9}$$

$$\sum_{b\in B}\Psi_b y_{bj}^{rail} + \sum_{d\in D}\lambda_d y_{do}^{joint} - \sum_{a\in A}\delta_a x_{aj} \ge 0, \forall j \in J \tag{10}$$

$$\sum_{c\in C}\chi_c y_{co}^{road} + \sum_{dd\in D}\lambda_d y_{do}^{joint} - \sum_{a\in A}\delta_a x_{aj} \ge 0, \forall j \in J \tag{11}$$

$$\delta_a, \Psi_b, \chi_c, \lambda_d \ge 0, \forall\ a \in A, b \in B, c \in C, d \in D \tag{12}$$

Equation (7) determines the minimal performance score for DMU o, where, similar to the optimistic model, the weighted input values are kept constant as outlined in Equation (8). Equations (9)–(11) enforce the condition that outputs must exceed inputs (i.e., Output - Input $\ge$0) to establish a minimum threshold of efficiency. Specifically, Equation (9) pertains to joint indicators (covering both rail and road), while Equation (10) focuses on the efficiency related to rail indicators. In contrast, Equation (11) addresses the efficiency values associated with road indicators. Finally, the decision variables are constrained by Equation (12).

We note that the pessimistic model is solved individually for each DMU (terminal) $o \in J$, resulting in the optimal value for each DMU $o \in J$, denoted as β_o^*.

4.1.3. Overall evaluation measure

After deriving the optimal optimistic value (α_o^*) and the pessimistic value (β_o^*) for each terminal $o \in J$, the overall sustainability score for that terminal can be calculated using Equation (13):

$$\sigma_o = \frac{\alpha_o^*}{\sqrt{\sum_{j\in J}\left(\alpha_j^*\right)^2}} + \frac{\beta_o^*}{\sqrt{\sum_{j\in J}\left(\beta_j^*\right)^2}} \tag{13}$$

It is recalled that α_o^* and α_j^* correspond to the optimal values of the objective function (1) for terminals $o \in J$ and $j \in J$, respectively.

Likewise, β_o^* and β_j^* represent the optimal values of the objective function (7) for terminals $o \in J$ and $j \in J$, respectively.

The overall score in Equation (13) provides a balanced measure of sustainability performance by capturing both optimistic and pessimistic perspectives. For further information about the double frontier NDEA model, please refer to Khanjarpanah et al. [85].

4.2. Multi-period bi-objective MIP model

In this section, we develop a multi-period bi-objective MIP model to represent the rail-road intermodal freight transport system. Subsequent sections will expand upon this model to mitigate disruption risks and handle epistemic uncertainties in both demand and supply.

Notations

Sets	
N	Set of all nodes in the network
P	Set of potential nodes for establishing new terminals
E	Set of existing terminal nodes
L	Set of all terminal nodes ($L = P \cup E$)
E_H	Set of nodes for highway stations
E_R	Set of nodes for railway stations
HR	Set of nodes for all highway and railway stations ($HR = E_H \cup E_R$)
O	Set of origins (suppliers)
D	Set of destinations (demand zones)
S	Set of disruption scenarios
T	Set of time periods
Parameterss	
f_j	Cost of constructing a terminal in location $j \in P$ ($)
m_j	Cost associated with mode switching at terminal $j \in L$ ($)
c_{ij}	Unit freight transportation cost from $i \in N$ to $j \in N$ ($)
d_i^t	Demand of zone $i \in D$ in period $t \in T$ (ton)
s_i^t	Supply amount of supplier $i \in O$ in period $t \in T$ (ton)
p_i^t	Penalty cost for unsatisfied demand of zone $i \in D$ in period $t \in T$ ($)
$\acute{c}_j$	Capacity of terminal $j \in L$ (ton)
$\acute{c}_{ij}$	Capacity of link (i,j), where $i \in N$ and $j \in N$ (ton)
e_j	GHG emissions related to opening new terminal in location $j \in P$ (g CO2eq)
$\acute{e}_j$	GHG emissions related to mode-switching operation at terminal $j \in L$ (g CO2eq)
e_{ij}	GHG emissions related to freight transportation from $i \in N$ to $j \in N$ (g CO2eq/ton)
b_{js}	Binary parameter equals to 1 if terminal $j \in E$ is disrupted under scenario $s \in S$, otherwise 0
F_s^*	The optimal cost for each disruption scenario $s \in S$ ($)
ρ	Robustness level
p_s	Probability of occurrence of scenario $s \in S$ ($0 \le p_s \le 1$)
M	A positive large number
g_j	Storing capacity of terminal j $\in L$ (ton)
h_j	Unit holding cost in terminal j $\in L$ (ton)
Variables	
V_j	1 if a terminal is constructed in location $j \in P$, otherwise 0
K	Number of established terminals (unit)
X_{js}^t	1 if mode-switching is performed at node $j \in P$ in period t under scenario $s \in S$, 0 otherwise
U_{is}^t	Quantity of lost demand for zone $i \in D$ in period $t \in T$ under scenario $s \in S$ (ton)
Y_{ijs}^t	Quantity of transported freight from node $i \in N$ to node $j \in N$ in period $t \in T$ under scenario $s \in S$ (ton)
W_{js}^t	Existing inventory in terminal $j \in L$ in period $t \in T$ under scenario $s \in S$ (ton)

4.2.1. Equations

$$min\, Z_1 = \sum_{j\in P} f_j V_j + \sum_{s\in S} p_s \left[\sum_{j\in L, t\in T} m_j X_{js}^t + \sum_{i,j\in N, t\in T} c_{ij} Y_{ijs}^t + \sum_{i\in D, t\in T} p_i^t U_{is}^t + \sum_{j\in L, t\in T} h_j . W_j^{ts} \right] \tag{14}$$

$$min\, Z_2 = \sum_{j\in P} e_j V_j + \sum_{s\in S} p_s \left[\sum_{j\in L, t\in T} \acute{e}_j X_{js}^t + \sum_{i,j\in N, t\in T} e_{ij} Y_{ijs}^t \right] \tag{15}$$

Subject to.

$$\sum_{j\in(i,j)\in E} Y_{ijs}^t \le s_i^t \forall i \in O, t \in T, s \in S \tag{16}$$

$$d_i^t \le \sum_{j|(j,i)\in E} Y_{jis}^t + U_{is}^t \forall i \in D, t \in T, s \in S \tag{17}$$

$$\sum_{j|(i,j)\in E_H} Y_{ijs}^t - \sum_{j|(i,j)\in E_H} Y_{jis}^t = 0 \forall\, i \in E_H, t \in T, s \in S \tag{18}$$

$$\sum_{j|(i,j)\in E_R} Y_{ijs}^t - \sum_{j|(i,j)\in E_R} Y_{jis}^t = 0 \forall\, i \in E_R, t \in T, s \in S \tag{19}$$

$$-M.X_{js}^t \le \sum_{i|(i,j)\in E_H} Y_{ijs}^t - \sum_{i|(j,i)\in E_H} Y_{jis}^t \le M.X_{js}^t \forall j \in L, t \in T, s \in S \tag{20}$$

$$-M.X_{js}^t \le \sum_{i|(i,j)\in E_R} Y_{ijs}^t - \sum_{i|(j,i)\in E_R} Y_{jis}^t \le M.X_{js}^t \forall j \in L, t \in T, s \in S \tag{21}$$

$$Y_{ijs}^t \le \acute{c}_{ij} \forall (i,j) \in E_N, t \in T, s \in S \tag{22}$$

$$\sum_j V_j = K \tag{23}$$

$$X_{js}^t \le V_j \forall j \in E, t \in T, s \in S \tag{24}$$

$$W_{js}^{t-1} + \sum_{i\in HR} Y_{ijs}^t \le \acute{c}_j V_j \forall j \in P, t \in T, s \in S \tag{25}$$

$$W_{js}^{t-1} + \sum_{i\in HR} Y_{ijs}^t \le \acute{c}_j (1 - b_{js}) \forall j \in E, t \in T, s \in S \tag{26}$$

$$W_{js}^t = W_{js}^{t-1} + \sum_{i\in HR} Y_{ijs}^t - \sum_{i\in HR} Y_{jis}^t \forall j \in L, t \in T, s \in S \tag{27}$$

$$\sum_{i\in HR} Y_{jis}^t \le g_j \forall j \in L, t \in T, s \in S \tag{28}$$

$$V_j, X_{js}^t \in \{0,1\} \forall j \in P, t \in T, s \in S \tag{29}$$

$$Y_{ijs}^t, U_{is}^t \in R^+ \forall i,j \in N, t \in T, s \in S \tag{30}$$

$$K \in Integer \tag{31}$$

Equation (14) aims to minimize the total expected cost, including the cost of establishing terminals, mode-switching at terminal nodes, transportation, unsatisfied demand, and holding costs in terminals. Equation (15) seeks to minimize the environmental impact of the intermodal network by reducing the expected total emissions. Constraint (16) ensures that supply restrictions are adhered to. Constraint (17) expresses the relationship between the transportation quantity of freight to each demand zone and the amount of unsatisfied demand. Constraints (18) and (19) specify that in a non-terminal intermediate node, there must be a balance between input and output flows for the highway and railway, respectively. Constraint (20) specifies that if the mode-switching state is activated at a terminal point, there is no guarantee of maintaining a balance between inflow and outflow freight quantity on the highway. Conversely, if the mode-switching state is not active at a terminal point, the inflow and outflow must be equal on the highway. Similar to the highway limitation in constraint (20), constraint (21) concerns the flow balance in the railway system at terminal nodes. Constraint (22) restricts the flow of each transportation link. Constraint (23) denotes the number of established terminals. Constraint (24) illustrates that if a terminal is established, then a mode-switching operation could be performed there. Constraint (25) indicates that if a potential terminal is established, a capacity limitation is imposed. Constraint (26) denotes the capacity of terminals, noting that if a terminal is disrupted in a disruption scenario,

it cannot be active. The balance constraint in terminal nodes is shown by constraint (27), and the output limitation in terminal nodes is depicted by constraint (28). Constraints (29)-(31) represent binary, positive, and integer variables, respectively.

4.3. P-robust method for managing disruption risks

Disruption risks can incur significant costs on transport networks if not adequately managed. To mitigate disruption risks and ensure acceptable network performance under varying scenarios, we employ the P-robust approach [88]. This method enhances traditional risk measures like regret and worst-case scenarios, proving more effective as recognized in studies. Its goal is to prevent total costs from becoming prohibitive under each disruption scenario [89].

In the P-robust method, let S denote the set of scenarios, $F^s(X,Y)$ represent the objective function value for each scenario $s \in S$, and F_s^* denote the minimal cost under scenario s. The method stipulates that the relative regret under each disruption scenario must not exceed a specified level, denoted as ρ, as shown in constraint (32):

$$\frac{F^s(X,Y)-F_s^*}{F_s^*} \le \rho \quad \forall s \in S \backslash \{1\} \tag{32}$$

Equation (32) measures the relative regret for scenario s, indicating the deviation of the objective function under disruption scenarios from F_s^*. Note that $s=1$ represents the scenario without disruption (nominal scenario). This equation can be linearly reformulated as Equation (33):

$$F^s(X,Y) \le (1+\rho)F_s^* \quad \forall s \in S \backslash \{1\} \tag{33}$$

Hence, by applying the P-robust method, the bi-objective model can be formulated as follows:

$$Min\ Z_1 = F^1(X,Y) = \sum_{j\in P} f_j V_j + \left[\sum_{j\in L,t\in T} m_j X_{j1}^t + \sum_{i,j\in N,t\in T} c_{ij} Y_{ij1}^t + \sum_{i\in D,t\in T} p_i^t U_{i1}^t + \sum_{j\in L,t\in T} h_j . W_{j1}^t \right] \tag{34}$$

Objective (15)
Subject to:

$$\sum_{j\in P} f_j V_j + \sum_{j\in L,t\in T} m_j X_{js}^t + \sum_{i,j\in N,t\in T} c_{ij} Y_{ijs}^t + \sum_{i\in D,t\in T} p_i^t U_{is}^t + \sum_{j\in L,t\in T} h_j . W_{js}^t \le (1+\rho)F_s^* \quad \forall s \in S \backslash \{1\} \tag{35}$$

Constraints (16) to (31)

Objective (34) aims to minimize the total cost under the nominal situation (when no disruption occurs). Constraint (35) is derived by substituting cost components in constraint (33) to ensure that the relative regret does not surpass the designated level ρ in any of the disruption scenarios.

4.4. PCCP for addressing epistemic uncertainty

To address epistemic uncertainty in demand and supply, we leverage the PCCP method. This method is well-known in fuzzy programming for effectively coping with possibilistic data and parameters [90]. PCCP utilizes two notable measures, namely possibility (*Pos*) and necessity (*Nec*) measures. A minimum confidence level (α and β) is set to handle the constraints that incorporate possibilistic data on either the left or right sides, or both. For a comprehensive explanation of the PCCP approach, please refer to Appendix A.

Applying the PCCP formulation from (A.14) to the optimization model outlined in Section 4.3, Equations (15)–(17) and (34) and (35)are adjusted as follows in Equation (36) through (40), respectively:

$$\begin{aligned} Min\ Z_1 = & \sum_{j\in P}\left[\frac{(1-\vartheta)}{2}f_j^1 + \frac{1}{2}f_j^2 + \frac{\vartheta}{2}f_j^3\right] V_j \\ & + \sum_{j\in L,t\in T}\left[\frac{(1-\vartheta)}{2}m_j^1 + \frac{1}{2}m_j^2 + \frac{\vartheta}{2}m_j^3\right] X_{j1}^t \\ & + \sum_{i,j\in N,t\in T}\left[\frac{(1-\vartheta)}{2}c_{ij}^1 + \frac{1}{2}c_{ij}^2 + \frac{\vartheta}{2}c_{ij}^3\right] Y_{ij1}^t \\ & + \sum_{i\in D,t\in T}\left[\frac{(1-\vartheta)}{2}p_i^{t1} + \frac{1}{2}p_i^{t2} + \frac{\vartheta}{2}p_i^3\right] U_{i1}^t \\ & + \sum_{j\in L,t\in T}\left[\frac{(1-\vartheta)}{2}h_j^1 + \frac{1}{2}h_j^2 + \frac{\vartheta}{2}h_j^3\right] W_{j1}^t \end{aligned} \tag{36}$$

$$min\ Z_2 = \sum_{j\in P}\left[\frac{(1-\vartheta)}{2}e_j^1 + \frac{1}{2}e_j^2 + \frac{\vartheta}{2}e_j^3\right] V_j + \sum_{s\in S} p_s \left(\sum_{j\in L,t\in T}\left[\frac{(1-\vartheta)}{2}\acute{e}_j^1 + \frac{1}{2}\acute{e}_j^2 + \frac{\vartheta}{2}\acute{e}_j^3\right] X_{js}^t + \sum_{i,j\in N,t\in T}\left[\frac{(1-\vartheta)}{2}e_{ij}^1 + \frac{1}{2}e_{ij}^2 + \frac{\vartheta}{2}e_{ij}^3\right] Y_{ijs}^t \right) \tag{37}$$

$$\begin{aligned} & \sum_{j\in P}\left[\frac{(1-\vartheta)}{2}f_j^1 + \frac{1}{2}f_j^2 + \frac{\vartheta}{2}f_j^3\right] V_j + \sum_{j\in L,t\in T}\left[\frac{(1-\vartheta)}{2}m_j^1 + \frac{1}{2}m_j^2 + \frac{\vartheta}{2}m_j^3\right] X_{js}^t \\ & + \sum_{i,j\in N,t\in T}\left[\frac{(1-\vartheta)}{2}c_{ij}^1 + \frac{1}{2}c_{ij}^2 + \frac{\vartheta}{2}c_{ij}^3\right] Y_{ijs}^t \\ & + \sum_{i\in D,t\in T}\left[\frac{(1-\vartheta)}{2}p_i^{t1} + \frac{1}{2}p_i^{t2} + \frac{\vartheta}{2}p_i^3\right] U_{is}^t \\ & + \sum_{j\in L,t\in T}\left[\frac{(1-\vartheta)}{2}h_j^1 + \frac{1}{2}h_j^2 + \frac{\vartheta}{2}h_j^3\right] W_{js}^t \\ & \le (1+\rho)F_s^* \quad s\in S\backslash\{1\} \end{aligned} \tag{38}$$

$$\sum_{j|(i,j)\in HR} Y_{ijs}^t \le \frac{(\alpha-\vartheta)s_i^{t1} + (1-\alpha)s_i^{t2}}{1-\vartheta} \quad i\in O, t\in T, s\in S \tag{39}$$

$$\sum_{j|(j,i)\in HR} Y_{ijs}^t + U_{is}^t \ge \frac{(\beta-\vartheta)d_i^{t3} + (1-\beta)d_i^{t2}}{1-\vartheta} \quad i\in D, t\in T, s\in S \tag{40}$$

4.5. TH method for solving the bi-objective model

To resolve the formulated model, we transform the bi-objective formulation into a single-objective model. An effective technique renowned for handling multi-objective problems is the TH method introduced by Torabi and Hassini [81]. This method necessitates defining two variables, namely the positive ideal solution (PIS) and negative ideal solution (NIS), for each objective function independently. Obtaining these solutions requires solving each objective function separately, with the optimal values corresponding to the PIS values. The NIS values can be derived using the following equations:

$$Z_1^{NIS} = Z_1\left(x_2^{PIS}\right) \tag{41}$$

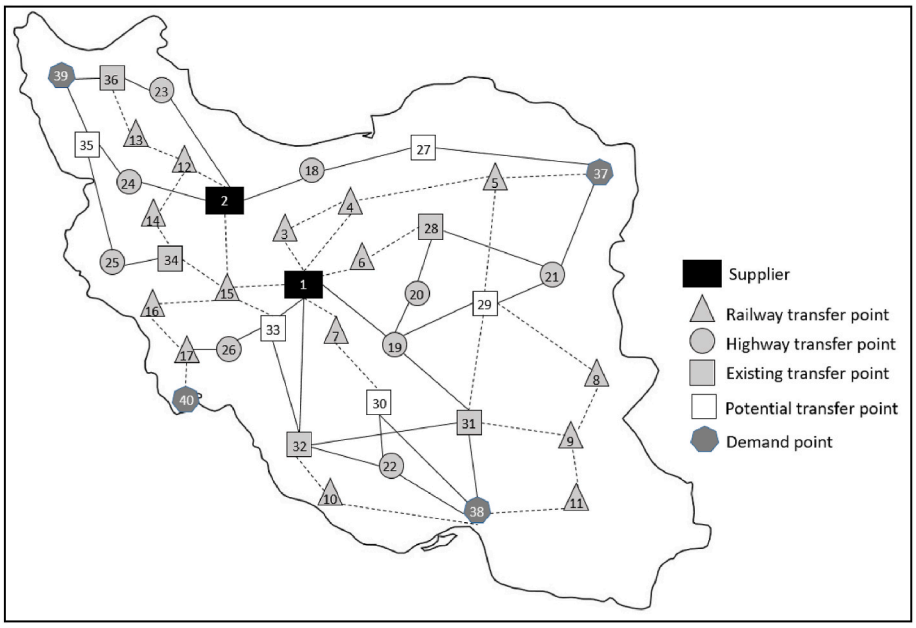


Fig. 4. The investigated rail-road intermodal freight transportation network.

$$Z_2^{NIS} = Z_2\left(x_1^{PIS}\right) \tag{42}$$

Once PIS and NIS are obtained, membership functions are formulated as follows:

$$\zeta_1(x) = \begin{cases} 1 & if\ Z_1 < Z_1^{PIS} \\ \dfrac{Z_1^{NIS} - Z_1}{Z_1^{NIS} - Z_1^{PIS}} & if\ Z_1^{PIS} \le Z_1 \le Z_1^{NIS} \\ 0 & if\ \ Z_1 \ge Z_1^{NIS} \end{cases} \tag{43}$$

$$\zeta_2(x) = \begin{cases} 1 & if\ Z_2 < Z_2^{PIS} \\ \dfrac{Z_2^{NIS} - Z_2}{Z_2^{NIS} - Z_2^{PIS}} & if\ Z_2^{PIS} \le Z_2 \le Z_2^{NIS} \\ 0 & if\ \ Z_2 \ge Z_2^{NIS} \end{cases} \tag{44}$$

The bi-objective formulation can then be transformed into a single-objective model, as expressed in Equations (45)–(47):

$$Max\ \Gamma(\mathrm{x}) = \lambda\beta_0 + (1-\lambda)\sum_h \theta_h \zeta_h(x) \tag{45}$$

$$s.t. \quad \beta_0 \le \zeta_h(x) \quad h = 1,2 \tag{46}$$

$$x \in G(x), \Gamma\ and\ \beta_0\ \in [0,1] \tag{47}$$

Here, $G(x)$ represents the feasibility domain, while θ_h and λ, represent the relative importance of each objective function and the balance of the TH method's objective function components. Furthermore, Γ the objective function value of the TH method, varies in $\in [0,1]$ interval. For further insights, one can refer to Torabi and Hassini [81].

Table 2
Input and output indicators of the double frontier NDEA for evaluation of intermodal terminals.

Measures	Inputs	Outputs		
		Rail	Road	Joint
Distance from main industrial zones	×			
Distance from airports	×			
Impact on the ecological landscape	×			
Risk of adverse meteorological condition	×			
Railway connectivity		×		
Roadway connectivity			×	
Distance from residential districts				×
Volume of international trade				×

Table 3
Parameter values in the case study.

Parameter	Value Range	Parameter	Value Range
c_{ij}	[20,60]	s_i^t	[1000,1200]
f_j	[2000,4000]	$\acute{e}_j$	[40,70]
h_j	[2,5]	e_{ij}	[20,50]
$\acute{c}_j$	[500,700]	α	0.9
$\acute{c}_{ij}$	[450,750]	β	0.9
d_i^t	[100,800]	ϑ	0.5

Table 4
Sustainability assessment summary using the double frontier NDEA for terminals.

DMU	Individual efficiency		Overall	Rank
	Optimistic	Pessimistic		
1	0.558	1	0.402	12
2	0.547	1	0.398	13
3	1	1.918	0.746	1
4	1	1.150	0.585	4
5	0.458	1	0.367	15
6	1	1	0.554	5
7	0.623	1	0.424	10
8	0.657	1	0.436	9
9	0.514	1	0.386	14
10	1	1	0.659	2
11	0.577	1.343	0.480	8
12	0.835	1.468	0.595	3
13	0.722	1.384	0.539	6
14	0.607	1	0.419	11
15	0.768	1.267	0.530	7

5. Case study and computational results

Using two sets of experiments, comprising a dataset from the literature and an actual freight transportation network, we validated the proposed methodology. We present the case study and sensitivity

Table 5
Inventory level in different periods.

Terminals	Scenario	Time periods				Terminals	Scenario	Time periods			
		1	2	3	4			1	2	3	4
27	10	–	–	87	–	27	14	–	–	180	–
27	15	–	–	180	–	27	21	–	–	180	–
27	22	–	–	180	–	27	24	–	–	38	–
27	26	–	–	180	–	27	27	–	–	38	–
27	32	–	–	38	–	28	1	100	100	100	–
30	3	–	386	–	–	30	4	–	386	140	–
30	8	–	386	140	–	30	11	–	386	140	–
30	12	–	57	180	–	30	14	–	460	–	–
30	15	–	170	–	–	30	17	–	460	–	–
30	18	–	227	–	–	30	19	–	170	–	–
30	20	–	420	140	–	30	21	–	170	–	–
30	22	–	170	–	–	30	23	–	460	142	–
30	24	–	170	142	–	30	27	–	460	142	–
30	28	–	170	130	–	30	30	–	170	142	–
30	31	–	170	130	–	30	32	–	170	142	–
31	1	100	100	–	–	32	1	100	86	56	–
32	2	–	386	140	–	32	3	–		140	–
32	5	–	386	140	–	32	6	–	170	180	–
32	7	–	386	140	–	32	9	–	170	180	–
32	10	–	170	93	–	32	12	–	403		–
32	13	–	170	180	–	32	16	–	170	180	–
32	18	–	233	180	–	32	19	–		180	–
32	25	–	170	180	–	32	26	–	170	–	–
32	29	–	170	180	–	34	1	100	100	–	–
35	17	–	–	180	–	35	23	–	–	38	–
35	28	–	–	50	–	35	30	–	–	38	–
35	31	–	–	50	–	36	1	100	74	24	–
36	2	–	74	40	–	36	3	–	74	40	–
36	4	–	74	40	–	36	5	–	74	40	–
36	7	–	74	40	–	36	8	–	74	40	–
36	9	–	74	40	–	36	11	–	74	40	–
36	20	–	40	40	–						

analyses in the following subsections along with computational results for the literature dataset in Appendix B. All numerical experiments were conducted on a laptop with a 2.2 GHz CPU and 8 GB of RAM, using the CPLEX solver in GAMS software to solve the mathematical models. Each numerical instance and the real-world case achieved optimality within 20 s.

5.1. Case description

We applied the proposed methodology to an actual rail-road freight transport network in the Middle East, aiming to redesign a resilient and sustainable intermodal freight system. This system is intended to transport food grain across the entire network. Fig. 4 illustrates the freight network under investigation.

Data for the case study were gathered from multiple sources, including historical records, expert interviews, and resources from consulting websites affiliated with the railway company.

To establish the sustainability criteria for our study, we followed a structured, three-step process to ensure that our criteria were grounded in both academic literature and regional expertise.

1. Literature Review: We conducted an extensive review of existing research on sustainable intermodal and rail-road freight transport, focusing on widely recognized sustainability metrics. This step enabled us to compile an initial list of criteria encompassing economic, environmental, and social dimensions, aligned with the principles of the triple bottom line [82].
2. Validation and Refinement through Expert Interviews: Next, we validated and refined these preliminary criteria through interviews with regional transport experts. This process provided valuable insights into region-specific challenges, regulatory standards, and infrastructure capabilities, allowing us to adjust and prioritize the criteria to better fit the Middle Eastern context.
3. Expert Panel for Final Selection: In the third step, a panel of experts was convened to evaluate and select the final set of criteria based on their relevance, feasibility, and alignment with regional sustainability goals.

Following this process, we identified eight key sustainability criteria.

- Distance from Main Industrial Zones: The proximity of intermodal terminals to industrial zones facilitates cost-effective and efficient cargo transport, particularly for perishable goods with limited lifespans [91].
- Distance from Airports: Although air transportation accounts for a smaller share of freight transport compared to rail and road, establishing intermodal terminals near airports can enhance cargo operations on an international scale [13].
- Impact on the Ecological Landscape: It is essential to consider the environmental impacts of opening terminals, including the potential destruction of regional natural resources and the ecological effects of land development [92].
- Risk of Adverse Meteorological Conditions: Harsh weather can hinder terminal operations [26], making areas with severe climates less desirable for intermodal terminals. For example, in the cold months, heavy snowfall in western and northwestern regions can disrupt rail and road transport.
- Railway Connectivity: This criterion evaluates the availability of nearby railroads and key rail stations to expedite transport processes. Establishing terminals near well-connected rail networks reduces loading and unloading costs and enhances the circulation of goods [13].
- Roadway Connectivity: Similar to railway connectivity, this criterion assesses the proximity of freeways and expressways to terminals, contributing to efficient transport processes [13].

Table 6
Mode-switching operations.

Terminals	Scenario	Time periods				Terminals	Scenario	Time periods			
		1	2	3	4			1	2	3	4
27	10	–	–	✓	✓	27	14	–	–	✓	✓
	15	–	–	✓	✓		21	–	–	✓	✓
	22	–	–	✓	✓		24	–	–	✓	✓
	26	–	–	✓	✓		27	–	–	✓	✓
	32	–	–	✓	✓						
28	1	✓	–	–	✓						
29	11	–	–	✓	–	29	17	–	–	✓	–
	20	–	–	✓	–		21	–	–	✓	–
	27	–	–	✓	–		28	–	–	✓	–
	31	–	–	✓	–		32	–	–	✓	–
30	1	✓	–	✓	–	30	2	✓	–	✓	–
	3	✓	✓	✓	–		4	✓	✓	✓	✓
	5	✓	–	✓	–		6	✓	–	✓	–
	7	✓	–	✓	–		8	✓	✓	✓	✓
	9	✓	–	✓	–		10	✓	–	✓	–
	11	✓	✓	✓	✓		12	✓	✓	✓	✓
	13	✓	–	✓	–		14	✓	✓	✓	✓
	15	✓	✓	✓	✓		16	✓	–	✓	–
	17	✓	✓	✓	✓		18	✓	✓	✓	–
	19	✓	✓	✓	–		20	✓	✓	✓	✓
	21	✓	✓	✓	✓		22	✓	✓	✓	✓
	23	✓	✓	✓	✓		24	✓	✓	✓	✓
	25	✓	–	✓	–		26	✓	–	✓	–
	27	✓	✓	✓	✓		28	✓	✓	✓	✓
	29	✓	–	✓	–		30	✓	✓	✓	✓
	31	✓	✓	✓	✓		32	✓	✓	✓	✓
31	1	✓	–	✓	–						
32	1	✓	✓	✓	✓	32	2	✓	✓	✓	✓
	3	✓	–	✓	–		5	✓	✓	✓	✓
	6	✓	✓	✓	✓		7	✓	✓	✓	✓
	9	✓	✓	✓	✓		10	✓	✓	✓	✓
	12	✓	✓	✓	–		13	✓	✓	✓	✓
	16	✓	✓	✓	✓		18	✓	✓	✓	✓
	19	✓	–	✓	✓		25	✓	✓	✓	✓
	26	✓	✓	✓	✓		29	✓	✓	✓	✓
34	1	✓	–	✓	–						
35	17	–	–	✓	✓	35	23	–	–	✓	✓
	28	–	–	✓	✓		30	–	–	✓	✓
	31	–	–	✓	✓						
36	1	✓	✓	✓	✓	36	2	–	✓	✓	✓
	3	–	✓	✓	✓		4	–	✓	✓	✓
	5	–	✓	✓	✓		7	–	✓	✓	✓
	8	–	✓	✓	✓		9	–	✓	✓	✓
	11	–	✓	✓	✓		17	–	✓	✓	–

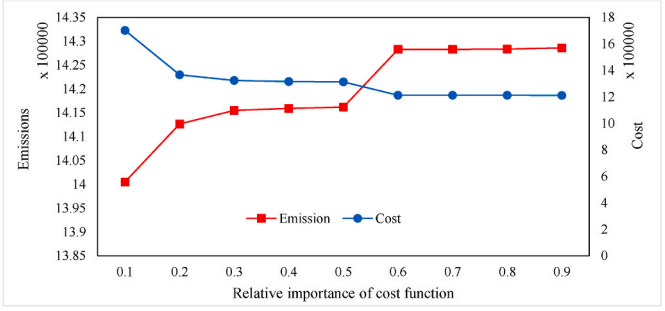


Fig. 5. Tradeoff between cost and emissions.

- Distance from Residential Districts: The distance from residential areas helps achieve social objectives by minimizing potential noise and traffic issues. Placing intermodal terminals in non-residential areas can mitigate impacts on nearby communities [91].

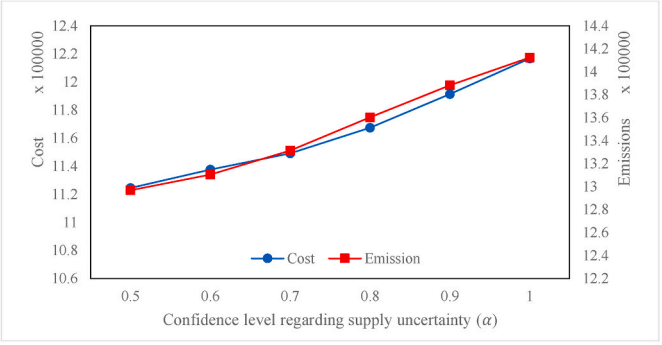


Fig. 6. Robustness analysis for supply uncertainty.

- Volume of International Trade: Regions with higher levels of international trade, such as those with significant manufacturing capabilities and export opportunities, are key factors in establishing

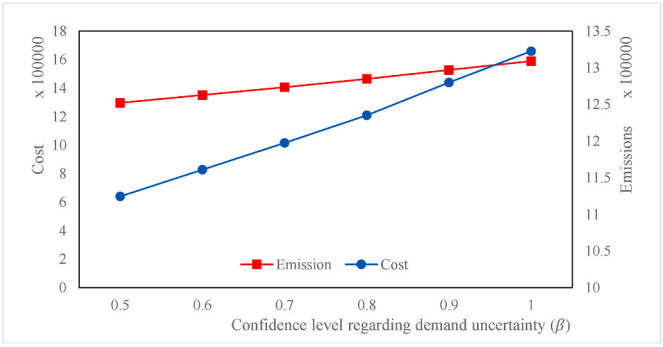


Fig. 7. Robustness analysis for demand uncertainty.

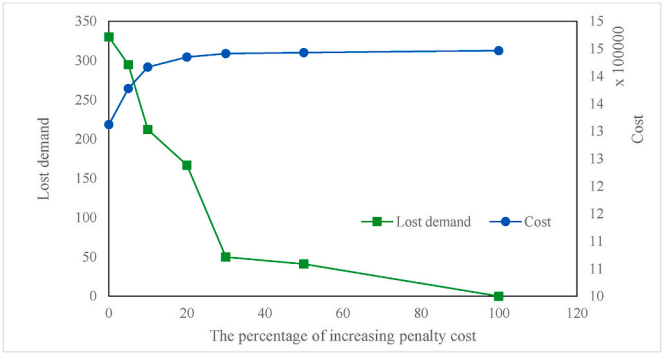


Fig. 8. The effect of penalty cost on the total cost and unmet demand.

intermodal terminals. Opening terminals near these areas enables efficient forwarding of goods to target markets [13].

To assess potential sites for intermodal terminals using the double frontier NDEA, the identified criteria were categorized into inputs and outputs. An indicator was classified as an output if a higher value was preferable, while it was considered an input if a higher value was less desirable. Table 2 summarizes the inputs and outputs considered for evaluating terminals using the double frontier NDEA.

Table 3 presents the parameter values used in this case study, detailing the range or specific values assigned to each parameter.

5.2. Implementation: optimal values for key decision variables

Assuming terminals as DMUs in the double frontier NDEA method, the overall sustainability performance of terminals were calculated. The results are summarized in Table 4. As is obvious from Table 4, DMU 3, 10, 12, 4, and 6 have the highest efficiency among the total DMUs, respectively. These DMUs correspond to terminals 27, 29, 30, 33, and 35 in Fig. 4. Assuming terminals as DMUs in the NDEA method, we calculated the overall sustainability performance of these terminals. The results are summarized in Table 4. As evident from Table 4, DMUs 3, 10, 12, 4, and 6 exhibit the highest efficiencies among all DMUs, respectively. These DMUs correspond to terminals 27, 29, 30, 33, and 35 in Fig. 4.

Tables 5 and 6 summarize inventory levels and mode-switching

Table 7
Cost and service level comparison between existing and redesigned systems.

Measure	Existing System	Redesigned Network	Improvement
Total cost	$1,558,611	$1,381,863	11.3 %
Maximum lost ratio	56.5 %	17.3 %	39.2 %
Average lost ratio	31.5 %	13.1 %	18.4 %

Table 8
Average inventory level comparison between existing and redesigned systems.

Terminals	Existing System	Redesigned Network	Change (%)
27	–	137.621	100
28	240.8	37.5	−84.4
29	–	–	–
30	–	915.981	100
31	28.1	25	−11.1
32	183	707.959	286.7
33	–	–	–
34	50	25	−50
35	–	44.487	100
36	96.0	148.951	55
Average	119.6	204.2	70.7

states across different periods and disruption scenarios. Table 5 shows that the redesigned network sustains a substantial inventory level, enhancing its capacity to meet demand and improve service levels. During disruptions, a significant portion of inventory-holding responsibilities is transferred to the newly opened Terminals 27 and 30.

Table 6 indicates that Terminal 30, along with Terminals 32 and 36, carries out the highest number of mode-switching operations within the redesigned network. Together, Tables 5 and 6 emphasize the essential roles of these terminals in maintaining higher inventory levels and more frequent mode-switching activities, underscoring their importance for the network's resilience and adaptability.

5.3. Tradeoff between cost and emissions

In investigating the interplay between total cost and emissions, we conducted a sensitivity analysis focusing on the significance of the cost objective function. Fig. 5 illustrates the outcomes, providing a practical tool for supply chain managers to navigate the tradeoff between cost efficiency and emissions reduction.

As expected, as the emphasis on economic aspects increases among supply chain managers, the optimization of the cost objective function improves, albeit with a rise in total emissions. What is particularly noteworthy is the variable steepness of these curves. Notably, when the importance parameter of the cost objective decreases from 0.6 to 0.5, we observe a significant reduction in emissions compared to other adjustments. This presents an opportune strategy for substantial emissions reduction without incurring considerable additional costs.

5.4. Robustness analysis

Figs. 6 and 7 depict how varying decision makers' confidence levels impact supply chain robustness against supply and demand uncertainties, respectively. As confidence levels rise, so does the supply chain's ability to withstand uncertainties, but this improvement comes with increased costs and emissions - the "price of robustness." These figures aid decision-makers in finding the right balance between robustness and its associated costs.

The price of robustness differs across these curves. For instance, transitioning from a confidence level of 0.6–0.7 in Fig. 6 has a smaller impact on total costs and emissions compared to shifting from 0.7 to 0.9. This suggests potential opportunities to enhance robustness at a lower cost within the supply chain.

An interesting observation emerges from comparing Figs. 6 and 7. The cost curve in Fig. 7 is steeper than that in Fig. 6, while the emissions curve is steeper in Fig. 6 than in Fig. 7. This indicates that building robustness against supply uncertainties leads to higher emissions, whereas robustness against demand uncertainties requires greater financial investment.

5.5. Tradeoff between cost and service level

We conducted a sensitivity analysis to better understand the correlation between total cost and unsatisfied customer demand, serving as a measure of service level. Fig. 8 illustrates the outcomes of this analysis, focusing on the variation in penalty cost associated with lost demand.

As depicted in Fig. 8, there is an upward trend in penalty cost leading to a reduction in lost demand, yet it also results in higher total costs. This emphasizes the delicate balance between financial considerations and service quality, where improving customer satisfaction by meeting demand necessitates careful consideration of these factors. A higher penalty cost signifies a more agile and responsive system, indicating that supply chain managers prioritize customer service even if it means incurring additional expenses. Fig. 8 is a helpful resource for supply chain managers, providing insights to help them navigate the tradeoff between the desired service level and total cost effectively.

5.6. Discussion and findings

This section presents a comparative analysis of the existing and redesigned networks, demonstrating cost and service improvements achieved through our optimization approach. Additionally, we provide managerial insights and practical implications to help supply chain managers enhance resilience, sustainability, and cost-efficiency.

5.6.1. Comparison with existing system

To gain a clearer understanding of the proposed optimization approach's impact in a real-world context, we compare the performance of the existing system with that of the redesigned network. The existing system reflects the current network prior to the implementation of the proposed approach, while the redesigned network represents the optimal configuration after applying the proposed approach. At a comparable environmental level, Table 7 summarizes our comparison of cost and service level metrics between the existing and redesigned systems. Additionally, Table 8 provides an analysis of inventory decisions across the two systems.

As shown in Table 7, the redesigned network achieves significantly better performance in terms of both cost and service level at the same environmental level. Cost savings are notably derived from the addition of new terminals, particularly Terminal 35 in the redesigned network. Service level metrics, such as maximum lost ratio (the highest proportion of unmet demand to total demand across disruption scenarios) and average lost ratio (the average proportion of unmet demand to total demand across scenarios), are also improved in the redesigned network through increased inventory levels. As illustrated in Table 8, the average inventory level rises by approximately 70.7 %, effectively enhancing resilience across various disruption scenarios.

5.6.2. Managerial insights and practical implications

Building on our findings, we present several key insights to guide managerial decision-making in optimizing supply chain resilience, sustainability, and cost-effectiveness.

- Our approach empowers supply chain managers to improve the sustainability of their transportation systems via intermodal solutions, introducing a resilient and sustainable method for mitigating disruptions and operational risks. This aligns with previous studies (e.g., Refs. [10,75]) that explore intermodal transport for sustainable freight solutions, though they did not account for operational uncertainties. By integrating intermodal strategies that consider uncertainties, managers can create a more resilient transportation network.
- By identifying efficient and inefficient terminals through double-frontier NDEA, as validated in previous studies within the energy sector (e.g., Ref. [85]), transport managers can make data-driven decisions regarding network design and operations. Terminals with higher rankings in our study, such as DMUs 3, 10, 12, 4, and 6, can be prioritized for establishment or capacity expansion. This approach enables targeted investments in high-performing areas, thereby enhancing the overall sustainability of the network.
- Higher average inventory levels, especially at strategic terminals, significantly strengthen the network's ability to meet demand during disruptions. The redesigned network shows a 70.7 % increase in average inventory, which acts as a buffer against shortages and improves service reliability. This observation aligns with recent literature advocating for resilient inventory management (e.g., Ref. [93]). Managers can strategically allocate inventory to ensure critical nodes are adequately stocked, thereby reducing vulnerability during unforeseen events.
- Frequent mode-switching at Terminals 30, 32, and 36 provides flexibility and adaptability, particularly during disruptions. This approach allows managers to adjust transportation modes dynamically based on real-time conditions, offering a practical way to respond to fluctuating demand, regulatory changes, or operational disruptions. Prioritizing terminals that can efficiently manage mode-switching helps maintain agility and responsiveness. It is essential to note that literature (e.g., Refs. [94,95]) highlights that these operations can increase emissions, which managers should consider by assigning appropriate importance to the emissions objective in the proposed mathematical model (see Equation (15)).
- Our results show that opening certain strategic terminals, particularly Terminal 35, significantly reduces network costs, which supports findings in freight transportation network optimization (e.g., Ref. [73]). Such targeted expansions allow for efficient cost management without overextending resources, supporting long-term network scalability.
- Sensitivity analysis indicates that a slight reduction in the cost-priority parameter—from 0.6 to 0.5—can yield significant emissions reductions without substantially increasing costs. This finding aligns with Demir et al. [36] on cost-environment trade-offs, showing that minor adjustments can lead to notable sustainability benefits. These trade-offs are particularly relevant for global transport companies aiming to balance profitability and carbon reduction goals.
- Our findings indicate that resilience measures to address supply uncertainties can raise emissions, which aligns with insights from Fahimnia and Jabbarzadeh [96]. Managers may consider offset policies or investments in clean technology at resilient nodes. Adopting low-emission technologies at critical inventory hubs can help manage emissions while supporting resilience goals, especially in sectors facing strict environmental regulations.
- Not all improvements in robustness have the same cost implications. For example, moving from a confidence level of 0.6–0.7 has a smaller cost and emissions impact than moving from 0.7 to 0.9. Managers can use these thresholds to make resilience investments that yield the highest returns. By prioritizing resilience adjustments with lower cost impacts, they can achieve robust performance while minimizing unnecessary expenditures, consistent with resilient supply chain network design literature (e.g., Refs. [96,97]).
- Higher penalty costs for unmet demand incentivize improvements in service levels. Managers can create a cost-effective, service-oriented transport system by balancing costs with service requirements. Our results show that minor cost increases can significantly improve service levels by reducing lost demand, aligning with findings on supply chain design (e.g., Ref. [22]).
- Consistent with resilient supply chain literature (e.g., Ref. [78]), our numerical experiments demonstrate the effectiveness of the proposed resilience strategy in mitigating disruption risks. For example, our analysis shows that implementing the P-robust approach can achieve a 24 % reduction in maximum regret with only a 1.4 % increase in expected costs. Additionally, our findings indicate that resilience strategies such as maintaining buffer inventory and

enabling mode-switching can deliver over 11 % in cost savings after adopting the proposed approach. Managers can leverage this framework to enhance resilience across various disruption scenarios by ensuring adequate inventory levels and transportation flexibility. These planning strategies are crucial for developing adaptive networks that can withstand demand and supply shocks, minimize potential losses, and enhance service consistency.

6. Conclusion

This paper presented a comprehensive multi-period optimization framework tailored for designing sustainable and resilient intermodal freight transport systems, offering an innovative logistics solution to address climate change challenges. Through the proposed framework, strategic decisions regarding facility location and tactical decisions concerning inventory and transport planning were effectively determined across several distinct phases. The initial phase involved a detailed analysis of the sustainability performance of intermodal terminals using a double frontier NDEA. Subsequently, a bi-objective MIP model was introduced in the second phase to simultaneously minimize total costs and carbon emissions. This model adeptly captured the intricacies of rail-road intermodal freight transport, including mode-switching operations, multi-period inventory management, and transport planning complexities. Further enhancing the model, the framework extended into addressing random disruptions and epistemic uncertainty through the integration of the P-robust method and PCCP. The final phase involved the implementation of the TH method to solve the bi-objective model, ensuring a green and efficient optimization process.

Validation was conducted using a literature-based dataset and a real rail-intermodal network, with results indicating substantial opportunities for supply chain managers to increase system sustainability and resilience through our proposed intermodal solutions. Some of the findings include the following.

- Opportunities exist to significantly reduce total emissions with only a modest increase in total costs. Sensitivity analysis demonstrated that lowering the cost-priority parameter from 0.6 to 0.5 leads to significant emissions reductions without substantial cost increases, equipping managers with effective strategies for carbon reduction.
- Small additional costs can yield significant enhancements in service levels and robustness against disruptions. The robustness analysis indicated that employing the P-robust method could achieve a 24 % reduction in maximum regret with just a 1.4 % increase in expected costs.
- The behavior of solutions revealed that strengthening robustness against supply uncertainties often results in higher emissions, while enhancing robustness against demand fluctuations may increase costs. These insights enable managers to make informed trade-offs, selecting resilience strategies that align with their environmental and cost priorities.
- Comparing existing systems with the redesigned system following the implementation of our optimization approach indicated potential cost savings exceeding 11 % while maintaining the same carbon footprint. A portion of this cost reduction was achieved through the establishment of sustainability-efficient and mode-switching terminals.
- The redesigned network demonstrated improved service levels, with a 22 % reduction in the maximum lost demand ratio and a 19 % decrease in the average lost demand ratio compared to the existing system. This improvement was realized by increasing average inventory levels at strategic terminals by 70.7 %, thereby enhancing the network's capacity to meet demand during disruptions.

Several limitations warrant further investigation. For instance, future studies could enrich sustainability assessments by incorporating a broader array of environmental, social and economic indicators. Additionally, while this work primarily emphasizes strategic and tactical planning, future research could include operational decision variables, such as vehicle routing and sequencing, to provide deeper insights into balancing economic and environmental performance.

CRediT authorship contribution statement

Shahab Rahiminia: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Amir Mehrabi:** Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Armin Jabbarzadeh:** Writing – review & editing, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Mohsen Pourseyed Aghaee:** Validation, Supervision, Methodology, Data curation, Conceptualization.

Acknowledgements

This research was supported by grants from The Natural Sciences and Engineering Research Council of Canada (NSERC), Fonds de Recherche du Québec – Nature et technologies (FRQNT), and Centre interdisciplinaire de recherche en opérationnalisation du développement durable (CIRODD).

Appendix A

Describing the PCCP method, let φ be a triangular fuzzy number presented as $\varphi = (\varphi_1, \varphi_2, \varphi_3)$ in which $\varphi_1 \le \varphi_2 \le \varphi_3$. The membership function for this fuzzy number can be written as follows:

$$\mu_x = \begin{cases} 0 \;\; \varphi_3 \le x, x \le \varphi_1 \\ 1 \quad x = \varphi_2 \\ \dfrac{x-\varphi_1}{\varphi_2-\varphi_1} \;\; \varphi_1 \le x \le \varphi_2 \\ \dfrac{\varphi_3-x}{\varphi_3-\varphi_2} \;\; \varphi_2 \le x \le \varphi_3 \end{cases} \tag{A.1}$$

After applying the possibility and necessity measures, the following equations are developed as the crisp form of fuzzy numbers for $\alpha \ge 0.5$:

$$Pos\{\varphi \le x\} \ge \alpha \leftrightarrow \frac{x-\varphi_1}{\varphi_2-\varphi_1} \ge \alpha \leftrightarrow \; x \ge (1-\alpha)\varphi_1 + \alpha\varphi_2 \tag{A.2}$$

$$Pos\{\varphi \ge x\} \ge \alpha \leftrightarrow \frac{\varphi_3-x}{\varphi_3-\varphi_2} \ge \alpha \leftrightarrow \; x \le (1-\alpha)\varphi_3 + \alpha\varphi_2 \tag{A.3}$$

$$Nec\{\varphi \le x\} \ge \alpha \leftrightarrow \frac{x-\varphi_1}{\varphi_2-\varphi_1} \ge \alpha \leftrightarrow \ x \ge (1-\alpha)\varphi_2 + \alpha\varphi_3 \tag{A.4}$$

$$Nec\{\varphi \ge x\} \ge \alpha \leftrightarrow \frac{\varphi_3-x}{\varphi_3-\varphi_2} \ge \alpha \leftrightarrow \ x \le (1-\alpha)\varphi_3 + \alpha\varphi_2 \tag{A.5}$$

The expected values of *Pos* and *Nec* can be calculated as follows:

$$E^{Pos}(\varphi) = \int_0^{+\infty} Pos(\varphi \ge x)dx - \int_{-\infty}^{0} Pos(\varphi \le x)dx = \frac{\varphi_2+\varphi_3}{2} \tag{A.6}$$

$$E^{Nec}(\varphi) = \int_0^{+\infty} Nec(\varphi \ge x)dx - \int_{-\infty}^{0} Nec(\varphi \le x)dx = \frac{\varphi_1+\varphi_2}{2} \tag{A.7}$$

Pos and *Nec* measures have an extreme view (optimistic and pessimistic) toward possibilistic data, which may rarely occur in the real world. To tackle this issue, Xu and Zhou [98] proposed the *Me* measure to enable a decision-maker to select a desired combination of *Pos* and *Nec* measures as follows:

$$Me(\varphi) = \vartheta.Pos(\varphi) + (1-\vartheta).Nec(\varphi) \tag{A.8}$$

A decision-maker's preferred combination can be obtained by varying ϑ in the interval [0, 1]. By setting $\vartheta = 0$, *Me* measure acts acts similarly to the *Nec* measure, and conversely, $\vartheta = 1$ leads to the *Pos* measure. For $\alpha \ge 0.5$, Xu and Zhou [98] indicated that *Me* measure can be formulated as follows:

$$E^{Me}(\varphi) = \int_0^{+\infty} Me(\varphi \ge x)dx - \int_{-\infty}^{0} Me(\varphi \le x)dx = \frac{(1-\vartheta)}{2}\varphi_1 + \frac{1}{2}\varphi_2 + \frac{\vartheta}{2}\varphi_3 \tag{A.9}$$

By setting ϑ less than 0.5, the model shows an inclination toward the *Nec* measure and adopts a pessimistic view, and the *Me* measure can be written as follows:

$$Me\{\widetilde{\varphi} \le x\} \ge \alpha \leftrightarrow \ \vartheta + (1-\vartheta)^* \frac{x-\varphi_2}{\varphi_3-\varphi_2} \ge \alpha \leftrightarrow x \ge \frac{(\alpha-\vartheta)\varphi_3 + (1-\alpha)\varphi_2}{1-\vartheta} \tag{A.10}$$

$$Me\{\widetilde{\varphi} \ge x\} \ge \alpha \leftrightarrow \ \vartheta + (1-\vartheta)^* \frac{\varphi_2-x}{\varphi_2-\varphi_1} \ge \alpha \leftrightarrow x \le \frac{(\alpha-\vartheta)\varphi_1 + (1-\alpha)\varphi_2}{1-\vartheta} \tag{A.11}$$

Using the above-mentioned concepts, we can now demonstrate how PCCP can be applied to an optimization model. Consider a general compact model where $\widetilde{F}$, $\widetilde{G}$, $\widetilde{B}$, and $\widetilde{E}$ indicate uncertain parameters subject to epistemic uncertainty, and X and Y indicate decision variables:

$$Min\ Z = \widetilde{F}x + \widetilde{G}y\ S.t..\ Ax \le \widetilde{B}\ Cx \ge \widetilde{E}\ Hy \le 1\ y \in \{0,1\}\ x \ge 0 \tag{A.12}$$

The PCCP formulation for the above model can be written as follows:

$$Min\ E[Z] = E[\widetilde{F}]x + [\widetilde{G}]y\ S.t..\ Me\{Ax \le B\} \ge \alpha\ Me\{Cx \ge E\} \ge \beta\ Hy \le 1\ y \in \{0,1\}\ x \ge 0 \tag{A.13}$$

According to Equations (A9)-(A11), model (A13) can be reformulated as follows (For more details about PCCP one can refer to Amiri et al. [99]):

$$Min\ E[Z] = \left[\frac{(1-\vartheta)}{2}F_1 + \frac{1}{2}F_2 + \frac{\vartheta}{2}F_3\right]x + \left[\frac{(1-\vartheta)}{2}G_1 + \frac{1}{2}G_2 + \frac{\vartheta}{2}G_3\right]y\ S.t..\ Ax \le \frac{(\alpha-\vartheta)B_1 + (1-\alpha)B_2}{1-\vartheta}\ Cx \ge \frac{(\beta-\vartheta)E_3 + (1-\beta)E_2}{1-\vartheta}\ Hy \le 1\ y \in \{0,1\}\ x \ge 0 \tag{A.14}$$

Appendix B

As outlined in the manuscript, we utilized two datasets to validate our proposed approach. The first dataset is an actual freight intermodal network, detailed in Section 5 of the manuscript. The second dataset is from Uddin and Huynh [59]. Figure B.1 visualizes the rail-road intermodal freight transport network considered in this dataset.

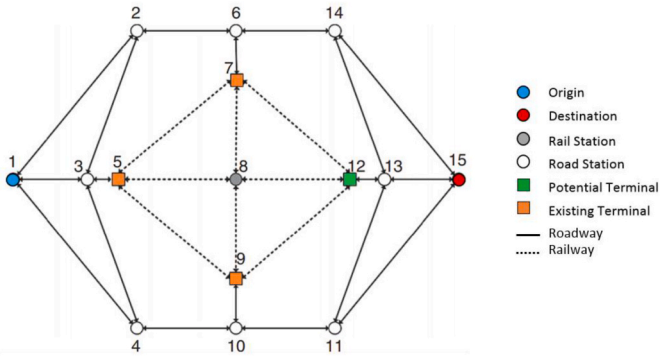


Fig. B.1. The rail-road intermodal freight transport network based on Uddin & Huynh's [59] data set.

Table B.1 provides a summary of results.

Table B.1
Summary of results for scenarios 1 and 8

Scenarios	Time periods					
	1		2		3	
1	Route (s)	1-2-6-14-15 1-4-10-11-15 1-3-5-7-12-13-15 1-2-6-7-12-13-15	Route (s)	1-2-6-14-15 1-4-10-11-15 1-3-5-7-12-13-15 1-3-5-9-12-13-15	Route (s)	1-2-6-14-15 1-4-10-11-15 1-3-5-7-12-13-15
	Lost (%)	8	Lost (%)	0	Lost (%)	0
8	Route (s)	1-2-6-14-15 1-4-10-11-15	Route (s)	1-2-6-14-15 1-4-10-11-15	Route (s)	1-2-6-14-15 1-4-10-11-15
	Lost (%)	16	Lost (%)	7	Lost (%)	21

Data availability

Data will be made available on request.

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